GRIT Track Badge interview scheduling

Write-up for the working prototype at **https://grit-web-production.up.railway.app** (one-tap demo sign-in for Student, Interviewer and Interview team; password for every demo account is `grit-demo-2026`). Stack: Supabase (Postgres, Auth, row level security, Realtime) and Railway.

1. States of a student's interview journey

State is kept per student per track and is derived in one database function from skills, bookings, outcomes and flags, so the student screen, the team table and the booking rules can never disagree.

State	The student is here when	What moves it on
Not eligible	Some required skills for the track are not cleared.	Last required skill contest cleared → **Eligible**.
Eligible, not booked	All skills cleared and no live booking. Also re-entered after a cancel, a retake, a first no-show, or when a cooldown ends.	Student confirms a slot → **Booked**.
Booked	A slot is held; join link opens 10 min before.	Student cancels or reschedules (until 12 h before) → Eligible / Booked. Interviewer or team cancels with a reason → Eligible. Slot end time passes → **Awaiting result**.
Awaiting result	Interview time is over, no outcome recorded yet.	Interviewer records Pass / Fail / Retake later / No-show.
Passed (badge earned)	Outcome is Pass. Badge and +500 points appear at once.	Outcome corrected within 24 h → matching state. Malpractice flag → **Nullified**.
Not cleared (cooldown)	Outcome is Fail. One attempt used; feedback shown.	14 days pass → Eligible. Third fail → **No attempts left**.
Retake later	Interview could not be completed (network, interviewer absent). No attempt used.	Immediately Eligible again, with a banner explaining why.
No-show	Student absent 10 min after the start.	First one: Eligible, no attempt used. Later ones count as an attempt.
No attempts left	3 attempts used this season.	Terminal for the season; team can override outside the tool.
Nullified	Team flagged malpractice, from any state.	Terminal for that track: badge removed, points removed, any upcoming interview cancelled, booking blocked. Appeal is outside the tool.

2. Edge cases handled, and assumptions

- **Two students, one slot, same instant.** The database lets exactly one through (section 3). The other sees the slot-taken message and fresh times; nothing is half-booked.
- **Double click or two tabs.** A second booking by the same student for the same track is rejected by a unique index, even if both requests arrive together.
- **Eligible for two tracks.** One live booking per track, and two bookings for the same student can't share a start time.
- **Late changes.** Students can reschedule or cancel until 12 h before; after that the buttons are disabled and the reason is shown.
- **Interviewer can't make it.** A booked slot can't be silently removed from availability. The interviewer cancels with a reason; the student is told, loses no attempt, and can rebook.
- **Outcome never recorded.** The slot shows "Needs outcome", the interviewer gets a banner, and the student appears under "Needs attention" for the team.
- **Wrong outcome recorded.** Editable for 24 h (pass can become fail and the badge is withdrawn, or the reverse). Every change is in the audit log with the previous value.
- **Malpractice flagged later.** Works from any state, including after a pass: badge and points are removed, upcoming interviews are cancelled and the slot is released.
- **Feedback versus private notes.** Students see feedback only. Private notes sit in a schema the API does not expose.
- **Students who have not booked.** Team view shows days since eligible, flags 7+ days, and logs reminders.
- **No capacity.** Team view compares open slots in the next 7 days with students still to book. Students with no open times get a clear empty state.
- **Time zones.** Stored in UTC, always shown in IST with the label.

Assumptions to confirm with the GRIT team (grit.niatindia.com was not reachable while building, so track names, skills and points are placeholders): 30-minute one-to-one interviews on Google Meet; students pick a time, not an interviewer, and any interviewer can assess any track; 14-day cooldown and 3 attempts per track per season; a Track Badge is worth 500 points and a nullified track contributes none; reminders go out by the existing GRIT notification channel (the prototype only logs them); appeals are handled by people, not the tool.

3. Data model

Table	Key fields	Notes
profiles	id (= auth user), role `student | interviewer | team`, full_name, campus	Role drives row level security.
tracks, track_skills	track id, name; skill id, track_id, name	Which skills a Track Badge needs.
student_skills	student_id, skill_id, cleared_at	Written by the contest system. Eligible = all of a track's skills present.
slots	id, interviewer_id, starts_at, ends_at, status `open | booked | blocked`	Unique (interviewer_id, starts_at). Interviewer availability.
bookings	id, slot_id, student_id, track_id, attempt_no, status `booked | completed | no_show | cancelled`, meet_url, cancelled_by, cancel_reason	History is kept: a cancel or reschedule never deletes a row.
outcomes	booking_id (PK), result `pass | fail | retake | no_show`, feedback, recorded_by, recorded_at, updated_at	One outcome per booking. Private notes live in `private.outcome_notes`.
track_standing	student_id + track_id (PK), badge_awarded_at, nullified_at	The badge itself. Nullified wins over everything.
malpractice_flags	id, student_id, track_id, reason, flagged_by, created_at	Team only.
audit_log	actor, action, student_id, track_id, detail (json), created_at	Bookings, changes, outcomes, reminders, nullifications.

How two students are stopped from booking the same slot. The browser never writes to tables. Every write is a database function, and reads are limited by row level security. Booking is one transaction with three layers:

```
-- 1. Lock one open slot at the chosen time (interviewers are pooled); a locked row is skipped, not waited on
select id from slots where starts_at = $1 and status = 'open' limit 1 for update skip locked;   -- none found → SLOT_TAKEN
-- 2. Flip it only if it is still open
update slots set status = 'booked' where id = $slot and status = 'open';                        -- 0 rows → SLOT_TAKEN
insert into bookings (slot_id, student_id, track_id, attempt_no) values (...);
-- 3. Backstops that hold even if application code is wrong
create unique index one_live_booking_per_slot on bookings (slot_id) where status in ('booked','completed','no_show');
create unique index one_open_booking_per_student_track on bookings (student_id, track_id) where status = 'booked';
```

The loser's transaction rolls back completely and they see: "That slot was just booked by someone else. Here are the nearest open times." Open times also update live on every screen through Supabase Realtime, so most students never reach the race at all.

Tested against the live database, with real sign-ins: 20 simultaneous bookings on one slot gave 1 booked, 19 rejected, 1 row. Four different students booking the last free slot at one time gave 1 booked, 3 rejected. One student firing two bookings at once gave 1 booked, 1 rejected. The interview team can rerun the first test from the **Race test** tab.

4. The three messages that matter most to a student

> **You're ready for your interview**
> You've cleared every skill for the Full Stack Badge. Book your interview to earn it. [Book interview]

> **Interview booked for Thu, 12 Nov, 4:00 PM IST**
> Your join link opens 10 minutes before. You can reschedule until Thu, 12 Nov, 4:00 AM.

> **You didn't clear the Full Stack interview this time**
> Your skills stay cleared, and you can book again from 26 Nov. *(followed by the interviewer's feedback)*

Why these three: the first is the only prompt that starts the flow, the second removes every "when, where, can I change it" question, and the third is the moment a student is most likely to give up, so it leads with what they keep and when they can try again. Pass, retake, cancelled-by-interviewer, slot-taken and nullified messages are also in the prototype.

5. Finding each part of the brief in the prototype

Booking: sign in as Aarav, Home → Book interview (then Reschedule or Cancel). **Availability and outcomes:** sign in as Priya, Availability and Schedule → Record outcome. **Status and result:** any student's Home; the login page has one-tap students in every state. **Booked versus not booked:** sign in as Sanjay, Students (filters, reminders, Flag malpractice), plus Interviews, Audit log and Race test. **Before and after:** the "Before and after" tab shows the home card, badge shelf, leaderboard row and public profile for pass, not cleared, retake and nullified.

Not in the prototype: real video calls, email or WhatsApp delivery of reminders, an appeals flow, and GRIT's real branding and track list.